S2M-Trek: From Single to Multi-Sphere Transport via Per-Frame Deep Sets on a Wheel-Legged Robot

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Abstract: We study the problem of scaling dynamic loco-manipulation from a single free-rolling sphere to multiple spheres transported simultaneously on the back of a wheel-legged quadruped, without fences, grippers, or mechanical stops. Multiple identical free-rolling spheres form an unordered set with no persistent identity: their ordering may change independently at each history frame, creating a *per-frame permutation symmetry* that standard history-concatenation set encoders do not explicitly enforce—these encoders impose only a shared, diagonal permutation symmetry over the full history. We show that this symmetry mismatch leads to a concrete failure mode in curriculum-based reinforcement learning. Within the same PPO training budget, flat MLPs and branch-wise encoders plateau at or below the two-sphere stage, while a history-concatenation Deep Sets baseline (DSHC) exhibits the same failure unless ball-to-slot assignments are randomized during training, suggesting that it exploits slot indices as a curriculum shortcut rather than learning identity-free multi-sphere dynamics. We propose **Per-Frame Deep Sets (PFDS)**, which performs permutation-invariant pooling within each history frame before temporal readout, achieving intrinsic invariance to independent per-frame permutations. A 2×2 ablation over encoder architecture and slot randomization separates the architectural and data-augmentation pathways to robust multi-sphere transport. Finally, we distill the PFDS teacher into TactSet via DAgger, replacing privileged sphere-state observations with a 16×16 Boolean union contact map, yielding a compact and naturally permutation-invariant tactile representation evaluated in simulation.

Keywords: locomotion, loco-manipulation, permutation invariance, deep sets, teacher-student learning, tactile sensing, multi-object manipulation

1 Introduction

Wheel-legged and legged robots have achieved remarkable locomotion versatility [1, 2, 3, 4], and recent work has extended these platforms to single-object loco-manipulation [5, 6, 7, 8]. Scaling from one object to many, however, is not a simple matter of adding more input channels. Transporting *multiple* free-rolling objects simultaneously—without fences, clamps, or fixed identity labels—changes the representational problem: the objects form an unordered, identity-free set whose slot assignments may change independently from one history frame to the next, a symmetry that prior work has not formalised. This paper studies the problem entirely in simulation using Isaac Lab [9]; real-robot deployment of the tactile student is in progress and will be reported separately.

The symmetry mismatch problem. Consider the multi-sphere observation tensor $X \in \mathbb{R}^{H\times N\times d}$, where H is the history length, N the maximum number of object slots, and d the per-object feature dimension. Because the spheres are identical, swapping two balls' slot assignments in any single history frame produces a physically equivalent state. The correct quotient group is therefore the

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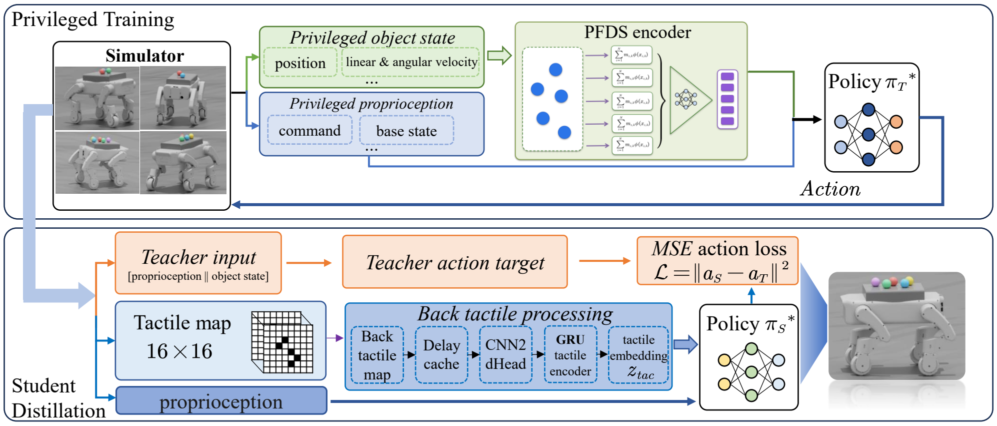

Figure 1: System overview (**top**: privileged teacher training; **bottom**: tactile-student distillation pipeline). The PFDS teacher is trained in Isaac Lab with PPO using privileged object-state observations (position, linear/angular velocity). TactSet is distilled via DAgger, replacing the privileged observation with a 16×16 Boolean union tactile contact map; physical deployment on the S3 robot is in progress and outside the scope of this paper.

per-frame product group

$$G_{\text{frame}} = (\mathfrak{S}_N)^H, \qquad (g \cdot X)_{t,i} = x_{t,\pi_t^{-1}(i)}, \quad g = (\pi_1, \dots, \pi_H), \tag{1}$$

not the *diagonal* subgroup $G_{\text{diag}} = \{(\pi, \dots, \pi) : \pi \in \mathfrak{S}_N\} \subsetneq G_{\text{frame}}$ that history-concatenation encoders [10, 11] actually satisfy. The quotient ratio $|G_{\text{frame}}|/|G_{\text{diag}}| = (N!)^{H-1}$ reaches $\approx 1.7\times10^6$ for $N{=}5$, $H{=}4$: six orders of magnitude of redundant non-physical variation that G_{diag} encoders cannot collapse.

Two paths to G_{frame} robustness. One can approach G_{frame} invariance in two ways:

1. **Architectural (intrinsic):** Design the encoder so that $f(g \cdot X) = f(X)$ for all $g \in G_{\text{frame}}$ by construction.
2. **Data-augmentation (learned):** Keep a G_{diag} encoder but sample a fresh per-frame slot permutation π_t at every simulation step, implicitly averaging the loss over the G_{frame} orbit.

We show experimentally that these two paths are not interchangeable: the architectural path advances the curriculum in both training regimes, while the data-augmentation path fails to advance past the two-sphere stage when augmentation is absent.

Contributions.

1. We formalize multi-sphere transport as a G_{frame}-invariant policy learning problem, deriving the quotient ratio $(N!)^{H-1}$ as a quantitative measure of the representational gap left unaddressed by standard encoders.
2. We propose **PFDS** (Per-Frame Deep Sets): a minimal modification of standard Deep Sets that achieves G_{frame}-invariance by performing frame-wise pooling before temporal readout, backed by proofs of invariance and universal approximation.
3. We present an eight-encoder comparison and a 2×2 ablation (encoder $\in \{\text{PFDS}, \text{DSHC}\}\times$ augmentation $\in \{\text{on}, \text{off}\}$) that separates the architectural and data-augmentation contributions. We further confirm PFDS's reliability across five random seeds; multi-seed runs of the baselines are left to future work.
4. We propose TactSet, a DAgger-distilled student that replaces privileged ball-state observations with a 16×16 Boolean union contact map—a representation that is *naturally* G_{frame}-invariant—achieving 75% no-drop transport of five spheres in simulation.

2 Related Work

Legged loco-manipulation. Rapid motor adaptation [4] and massively parallel RL [2] laid the foundation for agile locomotion. DribbleBot [5] and follow-up work [6, 7] extended quadrupeds to single-object manipulation, and LocoTouch [8] added back-mounted tactile sensing for cargo transport. To our knowledge, no prior work addresses unconstrained multi-sphere transport or identifies per-frame permutation symmetry as a representational bottleneck for scaling loco-manipulation.

Permutation-invariant networks. Deep Sets [10] introduced the $\rho(\sum_i \phi(x_i))$ universal approximator; Set Transformer [11] and PointNet [12] extended it with attention and point-cloud structure, and Maron *et al.* [13] analysed higher-order tensor invariance. These methods all target *single-frame* sets. We address the strictly harder *multi-frame* case, where each history frame may be permuted independently—a product group $(N!)^{H-1}$ larger than the diagonal group these encoders satisfy.

Equivariant RL and multi-object symmetry. MDP homomorphic networks [14, 15], SE(3)-equivariant manipulation [16, 17], and multi-agent symmetry methods [18] exploit spatial or agent-level structure for sample-efficient RL. None considers the temporal product symmetry arising when identical objects are observed over multiple history frames with independent slot re-assignments.

Tactile sensing and distillation. Tactile sensors support contact-rich manipulation and load transport on legged platforms [19, 20, 21, 22, 8], and DAgger [23] distillation bridges privileged training observations and deployment-feasible sensors. A central obstacle in multi-object settings is that object-indexed teacher representations do not map cleanly to tactile signals; we observe that the Boolean *union* of contact footprints is G_{frame}-invariant by construction and therefore aligns student and teacher representations without explicit permutation bookkeeping.

3 Problem Formulation

Multi-sphere transport MDP. The S3 robot—a self-designed four-wheel-independent-steering quadruped—transports $k \in \{1, \dots, N\}$ identical free-rolling spheres (radius $55\,\text{mm}$) on a flat back plate ($22.4\,\text{cm}\times15.1\,\text{cm}$). $N = 5$; the robot has 12 leg joints and 4 hub motors, controlled at $50\,\text{Hz}$. A back plate equipped with a 16×16 binary tactile array (256 cells) provides contact feedback. The episode terminates when any ball falls off the plate or the robot falls. Object count k is increased by a multi-criterion curriculum (episode-length ratio, support margin, dangerous fraction, edge-overflow fraction, velocity tracking error).

Multi-ball observation. At policy time t the agent observes a proprioception window $p_t \in \mathbb{R}^{d_p}$ (history length 6, see Section C.2) and a multi-ball history tensor $X_t \in \mathbb{R}^{H\times N\times d}$ (H=4 object frames, d=14: ball position, velocity, orientation, angular velocity, activity flag). Inactive slots are zero-masked. We use H exclusively for the object-history length throughout the main body, and write X for a generic such tensor when no time index is needed. Because the balls are identical, the correct physical equivalence class of a history tensor X is

$$[X]_{G_{\text{frame}}} = \{g \cdot X : g \in G_{\text{frame}}\},$$

and the policy value function should be constant on each orbit. We define the *curriculum-reachable ball count* at training budget B:

$$K_{\max}(E; B) = \max\big\{k : \mathrm{Succ}_k(E; B) \geq \eta\big\}, \tag{2}$$

where Succ_k is the success rate on k-ball evaluation episodes and η is the advancement threshold.

Evaluation metrics. We report two post-training success metrics for 100-episode evaluations; the 95% Wilson confidence interval is approximately $\pm$5–8 pp at success rates near 0 or 1 and widest ($\approx \pm 10$ pp) near 50%. *No-drop*: the episode completes all 500 steps without any sphere falling off the plate—this is the *primary* task metric, measuring whether the symmetry-aware encoder can maintain

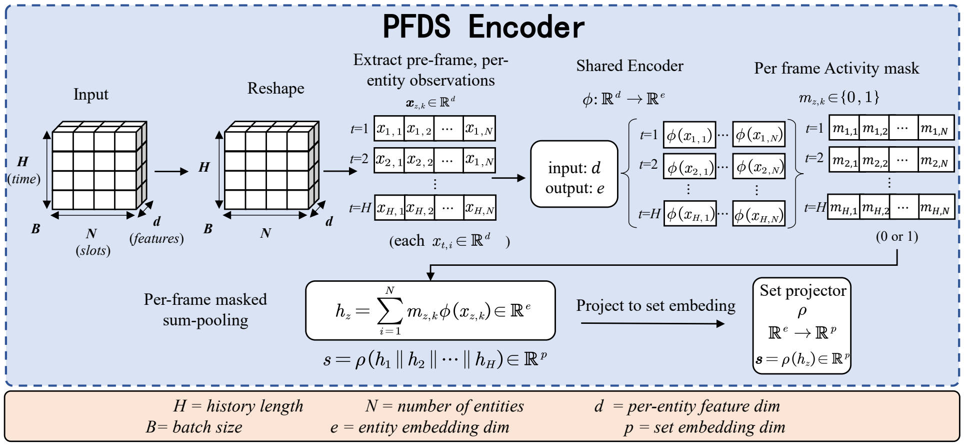

Figure 2: PFDS encoder architecture. Per-object observations $x_{t,i}$ are pooled independently within each history frame t to produce frame embeddings h_t; these are then concatenated and processed by the readout MLP ρ. Because pooling occurs *within* each frame, the encoder is invariant to independent per-frame permutations (G_{frame}-invariant by Proposition 1).

multi-sphere transport. *Strict*: additionally requires zero near-edge events throughout the episode (the ball never exits the safe zone), measuring conservative margin performance. Because near-edge events are sensitive to the amount of training received at each curriculum level, the strict metric conflates architectural quality with training exposure; we therefore report both but treat no-drop as the primary measure of encoder capability.

4 Method

4.1 Per-Frame Deep Sets (PFDS)

PFDS decouples *within-frame* set aggregation from *cross-frame* temporal fusion. It modifies standard Deep Sets [10] as

$$h_t = \mathrm{Pool}\big(\{\phi(x_{t,i})\}_{i\in\mathcal{A}_t}\big), \quad t = 1, \ldots, H, \tag{3}$$

$$z = \rho\big([h_1;\, h_2;\, \ldots;\, h_H]\big), \tag{4}$$

where $\phi : \mathbb{R}^d \to \mathbb{R}^e$ is a shared per-element MLP, Pool is masked sum, and ρ is a temporal readout MLP. A slot permutation π_t acts on both the observation tensor and the activity mask, mapping $(\mathcal{A}_t, x_{t,\cdot})$ to $(\pi_t(\mathcal{A}_t), x_{t,\pi_t^{-1}(\cdot)})$, so that h_t depends only on the multiset $\{x_{t,i}\}_{i\in\mathcal{A}_t}$ and is invariant to π_t for each t independently.

Proposition 1 (G_{frame}-invariance of PFDS)**.** *If* Pool *is permutation-invariant (e.g. sum, mean, max), then for all* $g = (\pi_1, \ldots, \pi_H) \in G_{\text{frame}}$, $f_{\text{PF}}(g \cdot X) = f_{\text{PF}}(X)$.

(Proof in Section A.)

Contrast with DSHC. The standard Deep Sets encoder [10] applied to multi-frame histories forms per-slot tokens $y_i = [x_{1,i}; \ldots; x_{H,i}] \in \mathbb{R}^{Hd}$ and computes

$$f_{\text{HC}}(X) = \rho\Big(\sum_{i=1}^{N} \phi_Z(y_i)\Big), \quad \phi_Z : \mathbb{R}^{Hd} \to \mathbb{R}^e. \tag{5}$$

We call this DSHC (Deep Sets, History-Concat). Because ϕ_Z couples frame index t with slot index i, f_{HC} satisfies only G_{diag} invariance, not G_{frame}. PFDS recovers G_{frame} invariance by pooling within each frame *before* temporal concatenation.

Proposition 2 (G_{diag} but not G_{frame} invariance of DSHC)**.** f_{HC} *is* G_{diag}*-invariant. However,* G_{frame}*-invariance fails in general: for* $H \geq 2$ *there exist a continuous feature map* ϕ_Z *(for instance* *the flattened outer-product feature $\phi_Z(y) = yy^\top$), a continuous readout ρ, and inputs X, $g \in G_{\mathrm{frame}} \setminus G_{\mathrm{diag}}$ such that $f_{\mathrm{HC}}(g \cdot X) \neq f_{\mathrm{HC}}(X)$.*

(Proof in Section A.)

The ratio $|G_{\mathrm{frame}}|/|G_{\mathrm{diag}}| = (N!)^{H-1}$ quantifies the residual representational gap; for $N{=}5,\ H{=}4$ this equals $(120)^3 \approx 1.7{\times}10^6$.

Theorem 1 (Universal approximation of PFDS for G_{frame}-invariant functions)**.** *Fix the activity pattern $\{\mathcal{A}_t\}_{t=1}^H$ so that each frame contains exactly $|\mathcal{A}_t|$ active slots, and let $\mathcal{X}$ be the compact quotient $\prod_{t=1}^H (K_t^{|\mathcal{A}_t|}/\mathfrak{S}_{|\mathcal{A}_t|})$ with $K_t \subset \mathbb{R}^d$ a compact box of admissible per-object observations. If $\phi : \mathbb{R}^d \to \mathbb{R}^e$ and $\rho : \mathbb{R}^{He} \to \mathbb{R}^a$ are realised by sufficiently wide continuous MLPs and* Pool *is a continuous permutation-invariant aggregator (sum or mean), then the family of functions representable by* PFDS *is dense in the space of continuous G_{frame}-invariant functions from $\mathcal{X}$ to $\mathbb{R}^a$ under the sup-norm.*

(Proof sketch in Section A; the result is a direct corollary of Deep Sets universality applied frame-wise followed by an MLP readout.)

4.2 Tactile Student Policy (TactSet)

The PFDS teacher uses privileged ball-state observations X_t during training. On the physical robot these are unavailable; only a $16{\times}16$ binary contact map is observed. TactSet bridges this gap by imitating the teacher from a tactile map $\tau_t \in \{0,1\}^{16\times16}$ defined as the Boolean union of active balls' contact footprints. Let $(\bar{x}_{t,i}, \bar{y}_{t,i}, \bar{z}_{t,i})$ denote the centre of ball i at time t expressed in the back-plate frame, and $(\bar{x}_s, \bar{y}_s)$ the planar centre of cell s in the same frame:

$$\tau_{t,s} = \bigvee_{i\in\mathcal{A}_t} \mathbf{1}\big(|\bar{x}_{t,i} - \bar{x}_s| < r + \delta_x,\ |\bar{y}_{t,i} - \bar{y}_s| < r + \delta_y,\ |\bar{z}_{t,i} - (z_{\mathrm{plate}} + r)| < \epsilon_z\big), \tag{6}$$

where z_{plate} is the height of the plate surface and the z-condition gates contact between the sphere's bottom pole and the plate. Because $\vee$ is symmetric in its arguments, τ_t depends only on the multiset of contact footprints and is a function of $[X_t]_{G_{\mathrm{frame}}}$, semantically aligning the student input with the teacher's observation. We stack tactile maps into a history $\mathcal{T}_t = [\tau_{t-H+1}; \ldots; \tau_t] \in \{0,1\}^{H\times16\times16}$ matching the object-history length H. The student minimises

$$\min_\theta\ \mathbb{E}_{(p_t, X_t, \mathcal{T}_t)\sim\mathcal{D}}\Big[\big\|\pi_S^\theta(p_t, \mathcal{T}_t) - \pi_T(p_t, X_t)\big\|_2^2\Big], \tag{7}$$

with DAgger-style online aggregation: behaviour-cloning initialisation followed by 7 rounds of online rollout with PFDS labelling. The student pre-encoder applies a 3×3 2-D CNN to the tactile history stack ($H{\times}16{\times}16$, 64 channels), and a GRU (hidden size 512) produces a 64-D tactile embedding that is concatenated with the proprioception embedding and fed to the policy head.

5 Experiments

5.1 Setup

All teacher policies are trained in Isaac Lab [9] with PPO [24] using 4096 parallel environments for 30 000 iterations. Reward, curriculum, and hyperparameters are identical across encoders; only the set-encoding module varies. Full hyperparameters are listed in Table 3.

Encoder baselines. We compare eight architectures spanning three symmetry levels (Table 1), all configured to comparable parameter budgets through matched hidden widths. G_0 *(no invariance):* Flat-MLP flattens X into a single vector; Branch-MLP encodes each slot $X_{:,i,:}$ with a shared MLP and concatenates the results. G_{diag} *invariance:* DSHC [10] forms history-concatenated tokens $y_i = [x_{1,i}; \ldots; x_{H,i}]$ and applies Deep Sets pooling; Set Transformer [11] replaces the inner network with ISAB/PMA attention on the same tokens. Their unaugmented variants (DSHC_OFF, Set

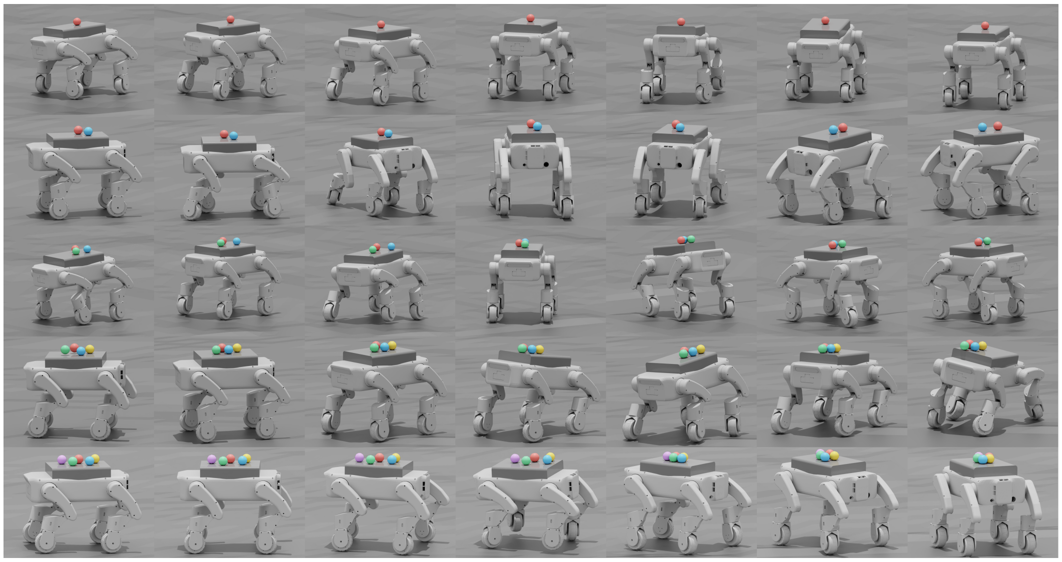

Figure 3: Qualitative snapshots of PFDS teacher transport in simulation. Each row shows frames from a representative episode at k=1 (top) through k=5 (bottom) balls.

Table 1: Teacher encoder comparison under equal PPO budget (30 000 iters). $K_{\max}$: highest curriculum level reached. "Adv. to k": iterations (thousands) to first reach that level; "—" = not reached within budget. Inv.: invariance group. Strict/no-drop: 5-ball success rates (100 trials, 95% Wilson CI ≈ ±5–8 pp); no-drop is the primary metric (§3). †: only G_{frame} encoder reaching k=5 without slot-permutation augmentation.

Encoder	**Inv.**	**perm**	$K_{\max}$	**Adv. to 3**	**Adv. to 5**	**strict**	**no-drop**
FLAT-MLP	G_0	T	≤ 1	—	—	0%	0%
BRANCH-MLP	G_0	T	≤ 2	—	—	0%	2%
DSHC_OFF	G_{diag}	F	≤ 2	—	—	0%	0%
DSHC	G_{diag}	T	5	∼7.4k	∼10.9k	68%	98%
Set Transformer	G_{diag}	T	5	∼3.6k	∼7.7k	10%	94%
Set Transformer (PF, ours)	G_{frame}	T	5	∼7.5k	∼16.4k	54%	100%
PFDS_OFF (ours)†	G_{frame}	F	5	∼16.9k	∼27.9k	41%	98%
PFDS (ours)	G_{frame}	T	5	∼3.9k	∼6.3k	41%	100%

Transformer without augmentation) expose the slot-identity shortcut by fixing slot assignments during training. G_{frame} *invariance (ours):* PFDS pools within each history frame before temporal readout (Proposition 1). Set Transformer (PF) is the higher-capacity analogue—frame-wise ISAB/PMA attention—but is not used as the distillation teacher because its wall-clock training time is the highest among G_{frame} encoders. PFDS_OFF disables slot randomisation for PFDS, isolating the architectural guarantee from the data-augmentation pathway.

5.2 Main Comparison: Architecture Determines Curriculum Reachability

Figure 5 reports curriculum progression and Fig. 4 reports per-encoder success rates. The results form a clear hierarchy consistent with symmetry-group containment: G_0 encoders (FLAT-MLP, BRANCH-MLP) stall at or below k=2, exposing a slot-index–ball-count shortcut; G_{diag} encoders (DSHC, Set Transformer) reach k=5 only with slot-permutation augmentation, and DSHC_OFF is blocked below k=3; G_{frame} encoders (PFDS, Set Transformer (PF)) advance under both regimes, the architectural guarantee that motivates our construction.

Primary metric (no-drop). PFDS and Set Transformer (PF) achieve 100% no-drop at k=5, the highest of any encoder, and PFDS is the fastest to reach the top curriculum level (∼6.3k iters to k=5, vs. ∼10.9k for DSHC and ∼7.7k for Set Transformer). The 2 pp no-drop gap over DSHC is within sampling noise at 100 trials; the substantive differentiator is that PFDS_OFF still attains 98% no-drop

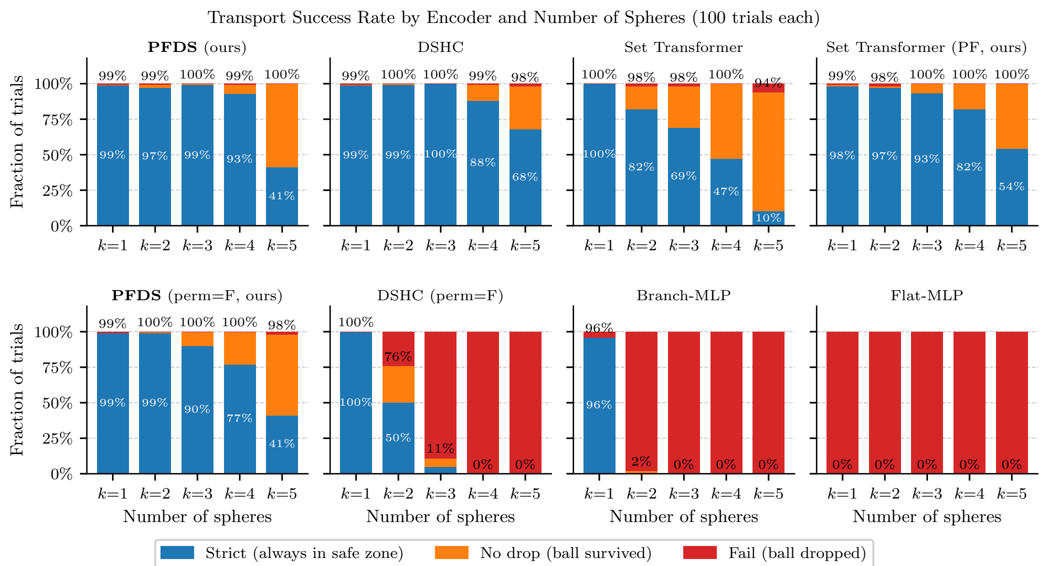

Figure 4: Transport success rates for all eight encoder architectures over 100 trials $\times$ 5 ball counts. Blue: strict (always within safe zone); orange: no-drop (ball survived, may briefly exit); red: fail. G_{frame}-invariant encoders (PFDS, Set Transformer (PF)) achieve 100% no-drop at k=5; DSHC with augmentation achieves 98%. G_0 encoders drop to near-zero no-drop beyond k=1; DSHC_OFF collapses beyond k=2.

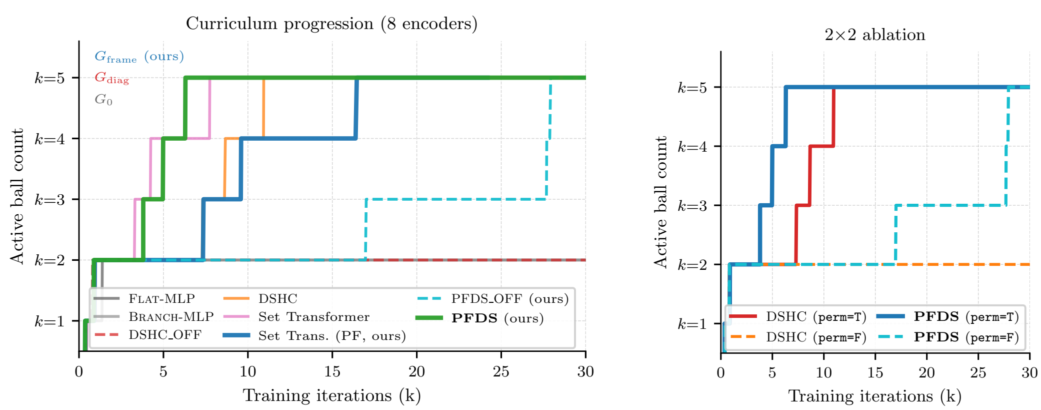

Figure 5: Training dynamics across encoder variants. **Left**: curriculum progression (active ball count vs. iteration) for all eight encoders. G_{frame} architectures (PFDS, Set Transformer (PF)) reach k=5 reliably; G_{diag} encoders only with slot-permutation augmentation; G_0 encoders stall at $k \leq 2$. **Right**: 2×2 ablation isolating the architectural and data-augmentation pathways. PFDS reaches k=5 with and without augmentation; DSHC fails without it.

without augmentation while DSHC_OFF does not advance past k=2 within budget. Robustness with a G_{frame}-invariant architecture is therefore independent of data augmentation, whereas with DSHC it is not.

Strict metric. DSHC reaches 68% strict at k=5 versus 41% for PFDS. This gap is not explained by five-ball training exposure: PFDS reaches k=5 about 4.6k iterations earlier and therefore accumulates *more*, not fewer, five-ball updates within the budget. With single-seed data we cannot separate policy-conservatism differences from seed variation, so we treat the gap as unexplained and leave matched-step, multi-seed comparisons to future work. No-drop captures the broader transport-success criterion and is the more direct test of the symmetry architecture.

Table 2: 2×2 ablation: encoder $\times$ per-step orbit augmentation. "Adv. to 3" is the number of training iterations at which the curriculum first reaches $k \geq 3$; "—" means not reached by 30 000 iterations.

Encoder	permute=True $K_{\max}$	permute=True Adv. to 3	permute=False $K_{\max}$	permute=False Adv. to 3
DSHC (G_{diag})	5	$\sim$7.4k	≤ 2	—
PFDS (G_{frame})	5	$\sim$3.9k	5	$\sim$16.9k

5.3 2×2 Key Ablation: Two Paths to G_{frame} Robustness

Table 2 and Fig. 5 confirm the theoretical prediction in our runs: DSHC reaches k=5 only when slot-permutation augmentation is applied at every step (sampling $\pi_t \sim \text{Uniform}(\mathfrak{S}_N)$ per environment per step) and does not advance past k=2 without it within the 30 000-iteration budget—the slot index becomes a spurious proxy for the curriculum stage that the G_{diag} encoder cannot ignore. PFDS succeeds in both settings: per-frame pooling structurally discards slot-identity information, so augmentation affects only the diversity of training states, not the invariance achieved. Section E gives mechanistic details.

5.4 Multi-Seed Robustness

To assess statistical robustness, we train five independent PFDS runs with seeds $\{40, 41, 42, 43, 44\}$ under identical PPO hyperparameters. All seeds converge to $k_{\max}$=5 within the 30 000-iteration budget; the $\pm 1\sigma$ band in Fig. 6 reflects variability in convergence timing rather than final performance. Multi-seed evaluation of baseline encoders is left to future work.

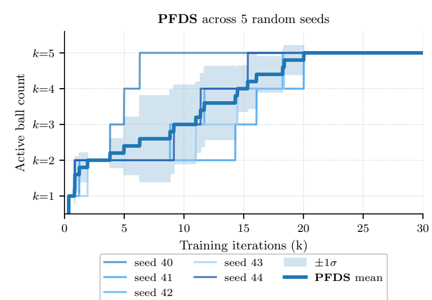

Figure 6: PFDS across five seeds (40–44). Mean (bold) $\pm 1\sigma$. All reach $k_{\max}$=5; variance is in timing only.

5.5 Tactile Student Distillation (TactSet)

We distill the PFDS teacher into TactSet via the DAgger protocol of Section 4.2 (400 000 BC steps + $7 \times 200\,000$ DAgger steps). The Boolean union contact map is itself G_{frame}-invariant, placing the student input in the same equivalence class as the teacher observation. Fig. 7 reports 100-trial evaluations of 500 steps per k: TactSet attains 97% no-drop and 92% strict at k=1, decreasing monotonically to 75% no-drop and 19% strict at k=5 (teacher: 100%/41% with privileged state). The 25 pp no-drop gap at k=5 has two plausible sources: (i) the union map is partially ambiguous near the plate boundary, and (ii) DAgger coverage is finite (8 rounds), under-labelling rare failure states. Additional DAgger rounds and state-entropy bonuses are plausible remedies that we leave to future work.

6 Conclusion

We introduced Per-Frame Deep Sets (PFDS), an intrinsically G_{frame}-invariant modification of Deep Sets for multi-sphere transport on a wheel-legged robot. Scaling loco-manipulation from one to five free-rolling objects requires aligning the encoder's symmetry group with the problem's true symmetry: the per-frame product group $G_{\text{frame}}=(\mathfrak{S}_N)^H$ is $(N!)^{H-1}$ times larger than the diagonal group G_{diag} that history-concatenation encoders satisfy. Our 2×2 ablation separates the architectural and data-augmentation pathways to G_{frame} robustness, and the eight-encoder comparison shows that G_{frame}-invariant architectures are the only encoders that remain robust without slot-permutation augmentation; among augmented encoders, no-drop differences at k=5 lie within the 100-trial sampling noise. Five-seed evaluation

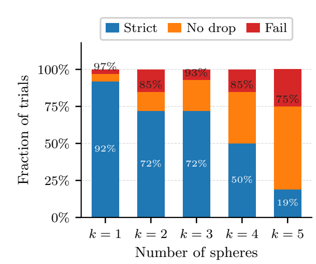

Figure 7: TactSet evaluation (100 trials of 500 steps per k). Bar-top: no-drop; inside blue bars: strict.

confirms that PFDS reaches $k_{\max}=5$ across all five of our runs. TACTSET shows that the privileged-state policy can be distilled into a purely tactile student that inherits permutation invariance from the Boolean union contact map.

Limitations and future work. All reported numbers are simulation results in Isaac Lab; transfer to the physical S3 robot (sensor noise, actuator delay, dynamics mismatch) is in progress and will be reported separately. Our 100-trial evaluations yield Wilson confidence intervals of $\pm$5–8 pp near the extremes, which suffice for the qualitative trends we report but not for fine-grained comparisons. Multi-seed runs for the DSHC and Set Transformer (PF) baselines, and a matched-steps-at-k=5 comparison that would resolve the strict-rate confound, are left to future work. Beyond the current setting, the per-frame pooling principle could be adapted to typed or heterogeneous object sets, and to tasks where slot identity between frames is unstable, by composing per-type Deep Sets blocks; we leave these extensions to future work.

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A Proof Details

A.1 Proof of Proposition 1 (PFDS is G_{frame}-Invariant)

Proof. The action of $g = (\pi_1, \ldots, \pi_H) \in G_{\text{frame}}$ maps slot i to $\pi_t^{-1}(i)$ at frame t and the active set $\mathcal{A}_t$ to $\pi_t(\mathcal{A}_t)$. Therefore

$$h_t(g \cdot X) = \mathrm{Pool}\big(\{\phi(x_{t,\pi_t^{-1}(i)})\}_{i \in \pi_t(\mathcal{A}_t)}\big) = \mathrm{Pool}\big(\{\phi(x_{t,j})\}_{j \in \mathcal{A}_t}\big) = h_t(X),$$

where the second equality reindexes the multiset by $j = \pi_t^{-1}(i)$, which traverses $\mathcal{A}_t$ exactly once. Because Pool is permutation-invariant the multiset value is unchanged. Hence $z(g \cdot X) = \rho([h_1; \ldots; h_H]) = z(X)$. $\square$

A.2 Proof of Proposition 2 (DSHC is not G_{frame}-Invariant)

We give an $H = 2$, $N = 2$ counterexample. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with rows indexing time and columns indexing slots, and take $g = (\mathrm{id}, (12)) \in G_{\text{frame}} \setminus G_{\text{diag}}$. Then $(g \cdot X)_{1,:} = (a, b)$ and $(g \cdot X)_{2,:} = (d, c)$, so the DSHC tokens are $y_1^g = (a, d)$, $y_2^g = (b, c)$, whereas $y_1 = (a, c)$, $y_2 = (b, d)$. For the linear feature $\phi_Z = \mathrm{id}$, $\sum_i y_i^g = (a + b, c + d) = \sum_i y_i$, so the linear feature happens to be invariant. For the flattened outer-product feature $\phi_Z(y) = yy^\top$, the off-diagonal $(1, 2)$ entry of $\sum_i \phi_Z(y_i^g)$ equals $ad + bc$, while the same entry of $\sum_i \phi_Z(y_i)$ equals $ac + bd$; these differ whenever $a \neq b$ and $c \neq d$. Choose ρ to project onto the $(1, 2)$-entry coordinate of its input. Then $f_{\text{HC}}(g \cdot X) - f_{\text{HC}}(X) = (ad + bc) - (ac + bd) \neq 0$, exhibiting the required ϕ_Z, ρ, X, and $g \in G_{\text{frame}} \setminus G_{\text{diag}}$. $\square$

A.3 Proof Sketch of Theorem 1 (Universal Approximation of PFDS)

Fix the per-frame cardinalities $|\mathcal{A}_t|$ and take a common compact box $K \supseteq \bigcup_t K_t \subset \mathbb{R}^d$. The set of admissible per-frame multisets is the quotient $K^{|\mathcal{A}_t|}/\mathfrak{S}_{|\mathcal{A}_t|}$ with the quotient topology induced by the product topology; this topology preserves multiplicity, unlike the Hausdorff metric. By the Deep Sets universality theorem [10] applied to multisets of size up to $\max_t |\mathcal{A}_t|$, there exists a single continuous $\phi : \mathbb{R}^d \to \mathbb{R}^M$ such that, for every t, the sum-pooling map $\Phi_t(S) = \sum_{i \in S} \phi(x_i)$ is a continuous injection from $K^{|\mathcal{A}_t|}/\mathfrak{S}_{|\mathcal{A}_t|}$ into $\mathbb{R}^M$. The same ϕ is therefore shared across frames, matching the PFDS architecture. Concatenating frame-wise, $v(X) = [\Phi_1(X_1); \ldots; \Phi_H(X_H)] \in \mathbb{R}^{HM}$, yields a continuous injection from $\mathcal{X}$ into $\mathbb{R}^{HM}$ whose fibres are exactly the G_{frame}-orbits, because two elements share the same v iff they share the same per-frame multiset for every t. Since $\mathcal{X}$ is compact and v is continuous, $v(\mathcal{X})$ is compact. Any continuous G_{frame}-invariant function $f : \mathcal{X} \to \mathbb{R}^a$ factors uniquely through v as a continuous $\tilde{f} : v(\mathcal{X}) \to \mathbb{R}^a$, and the standard universal approximation theorem for MLPs yields a ρ approximating $\tilde{f}$ to arbitrary sup-norm precision. $\square$

B Hyperparameters

C Environment Specification

C.1 Reward Function

The total reward r_t is the sum of object-balancing terms, locomotion terms, and a penalty term:

$$r_t = r_{\text{obj}} + r_{\text{loco}} + r_{\text{penalty}}. \tag{8}$$

Object-balancing rewards. Let $\mathcal{A}_t \subseteq \{1, \ldots, N\}$ be the set of active ball indices at time t. For each ball $i \in \mathcal{A}_t$, define the *support margin* $\mu_i \in [0, 1]$ as the minimum normalized distance from

Table 3: PPO hyperparameters shared across all teacher encoder experiments.

Parameter	**Value**	**Note**
Parallel environments	4096	Isaac Lab
Total iterations	30 000	
Steps per env per iter	24	
Mini-batches	4	
PPO epochs per iter	5	
Learning rate	3×10^{-4}	Adam
Discount γ	0.99	
GAE λ	0.95	
Clip range ϵ	0.2	
KL target	6×10^{-3}	
Entropy coeff.	0.01	
Value loss coeff.	1.0	
Max grad norm	1.0	
Control frequency	50 Hz	
History length H	4	
Max ball count N	5	

Table 4: DAgger distillation hyperparameters (TACTSET).

Parameter	**Value**
Rounds	8
BC steps (round 0)	400 000
DAgger steps (rounds 1–7)	200 000 each
Batch size (training)	20 000
Teacher inference batch	2 048
Learning rate	5×10^{-4}
CNN kernel	3×3, 64 channels
GRU hidden size	512
Tactile input shape	$(H, 16, 16)$
CNN embedding dim	64

ball i's projected position to the plate boundary (1 = plate center, 0 = edge):

$$r_{\text{margin}} = +3.0 \ \cdot \ \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \mu_i, \tag{9}$$

$$r_{\text{tail}} = +1.25 \ \cdot \ \mathrm{CVaR}_{0.4}\big(\{\mu_i\}_{i \in \mathcal{A}_t}\big), \tag{10}$$

where $\mathrm{CVaR}_{0.4}$ is the conditional mean of the lowest 40% of margins (a tail-risk reward term incentivizing stabilization of the most precarious ball).

$$r_{\text{edge}} = -15.0 \ \cdot \ \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \mathbf{1}[\mu_i < 0], \tag{11}$$

$$r_{\text{spacing}} = +1.5 \ \cdot \ \frac{2}{|\mathcal{A}_t|(|\mathcal{A}_t| - 1)} \sum_{i<j} \min\big(\|p_i - p_j\|_2,\ d_{\max}\big), \tag{12}$$

$$r_{\text{spacing,tail}} = +0.75 \ \cdot \ \mathrm{CVaR}_{0.4}\big(\{\|p_i - p_j\|_2\}_{i<j}\big), \tag{13}$$

$$r_{\text{dangerous}} = -20.0 \ \cdot \ \mathbf{1}\big[\min_{i \in \mathcal{A}_t} \mu_i < \mu_{\text{thresh}}\big], \tag{14}$$

where $p_i \in \mathbb{R}^2$ is the ball's projected position on the plate, $d_{\max}$ is a clipping radius, and μ_{thresh} is the dangerous-state threshold (set per curriculum level).

Velocity-related penalties.

$$r_{\text{vel},xy} = -0.05 \ \cdot \ \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \|v_i^{xy}\|_2^2, \tag{15}$$

$$r_{\text{vel},z} = -0.25 \ \cdot \ \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} |v_i^z|, \tag{16}$$

where v_i^{xy} and v_i^z are the ball's horizontal and vertical velocity in the robot frame.

Locomotion rewards (from base locomotion policy).

$$
\begin{aligned}
r_{\text{track},v} &= +1.5 \ \cdot \ \exp\bigl(-\|\hat{v}^{xy} - v_{\text{cmd}}^{xy}\|_2^2/0.25\bigr), && (17)\\
r_{\text{track},\omega} &= +1.25 \ \cdot \ \exp\bigl(-|\hat{\omega}_z - \omega_{z,\text{cmd}}|^2/0.25\bigr), && (18)\\
r_{\text{height}} &= -1.0 \ \cdot \ (h - h_{\text{ref}})^2, && (19)\\
r_{\text{roll,pitch}} &= -1.25 \ \cdot \ \|\theta_{\text{rp}}\|_2^2, && (20)
\end{aligned}
$$

where $\hat{v}^{xy}$ and $\hat{\omega}_z$ are the robot's base velocity, v_{cmd}^{xy} and $\omega_{z,\text{cmd}}$ are the commanded velocities, h is the base height, h_{ref} is the target height, and θ_{rp} are the roll and pitch angles.

C.2 Observation Space

The teacher policy receives the following observations at each timestep:

Proprioception (history length $H_{\text{prop}} = 6$, distinct from the object-history length $H = 4$).

- Base angular velocity $\omega_b \in \mathbb{R}^3$
- Projected gravity vector $g_b \in \mathbb{R}^3$
- Velocity command $[v_x, v_y, \omega_z] \in \mathbb{R}^3$
- Joint positions $q \in \mathbb{R}^{12}$ (relative to default)
- Joint velocities $\dot{q} \in \mathbb{R}^{12}$
- Previous action $a_{t-1} \in \mathbb{R}^{12}$
- Robot base height $h \in \mathbb{R}^1$

Total proprioception dimension: $43 \times H_{\text{prop}} = 258$.

Multi-ball state (object-history length $H = 4$, per-ball feature dim $d = 14$). For each active ball $i \in \mathcal{A}_t$:

- Relative position $\Delta p_i \in \mathbb{R}^3$ (ball position minus robot base, in robot frame; scale 1.0)
- Relative linear velocity $\Delta v_i \in \mathbb{R}^3$ (scale 0.5)
- Relative orientation as quaternion $q_i \in \mathbb{R}^4$ (scale 1.0)
- Relative angular velocity $\omega_i \in \mathbb{R}^3$ (scale 0.25)
- Activity flag $a_i \in \{0, 1\}$ (scale 1.0)

Inactive slots are zero-masked. Observation noise (uniform): position ± 0.01 m, velocity ± 0.2 m/s, orientation ± 0.05 (Euler-angle equivalent), angular velocity ± 0.2 rad/s. Ball state is observable only when the ball is in contact with the robot surface (contact detection threshold $\approx 10^{-8}$ N); otherwise the observation is zero.

The critic (teacher) additionally receives the denoised version of the above (no observation noise added).

C.3 Termination Conditions

An episode terminates (non-timeout) when any of the following occurs:

1. **Ball falls below robot base**: any active ball's z-position drops below the robot base z-position, $\exists i \in \mathcal{A}_t : p_{i,z} < p_{\text{robot},z}$.
2. **Robot falls**: the projected gravity vector indicates a roll angle $|\arcsin(g_{b,y})| > 90°$.
3. **Base height too low**: the robot base height falls below the minimum standing height threshold.

Note: the `base_contact` termination (body contact with the ground) is disabled for the multi-sphere task to prevent false positives from the tactile plate surface.

D Curriculum Design

D.1 Promotion Criteria

The multi-ball curriculum promotes from k to $k+1$ active balls when a rolling window of recent episodes satisfies *all six* of the following conditions simultaneously:

1. **Episode-length ratio** ≥ 0.85: the mean episode length divided by the maximum episode length exceeds 85%, indicating the agent consistently survives to episode end.
2. **Support margin** $\geq \mu_k^*$: mean minimum support margin across active balls exceeds the level threshold.
3. **Dangerous fraction** $\leq f_{d,k}^*$: the fraction of timesteps with any ball in a dangerous state (near edge) is below the level threshold.
4. **Edge-overflow fraction** $\leq f_{e,k}^*$: the fraction of timesteps with any ball outside the plate boundary is below the level threshold.
5. **Linear velocity tracking error** $\leq \epsilon_{v,k}^*$: mean $\|\hat{v}^{xy} - v_{\text{cmd}}^{xy}\|_2$ is below threshold.
6. **Angular velocity tracking error** $\leq \epsilon_{\omega,k}^*$: mean $|\hat{\omega}_z - \omega_{z,\text{cmd}}|$ is below threshold.

All conditions must hold simultaneously for `required_successes` $= 1$ consecutive evaluation windows, with `min_stage_episodes` $= 2048$ and `min_steps_per_level` $= 3200$.

D.2 Per-Level Promotion Thresholds

Table 5: Per-level curriculum promotion thresholds ($k \to k+1$). "Support margin" μ^* is the normalized minimum distance from plate boundary. "Dangerous frac." f_d^* and "Edge frac." f_e^* are allowed fractions of unsafe timesteps. "Vel. err." ϵ_v^* and "Yaw err." ϵ_ω^* are maximum allowed tracking errors.

Level ($k \to k+1$)	μ^*	f_d^*	f_e^*	ϵ_v^*	ϵ_ω^*
$1 \to 2$	0.30	0.20	0.02	0.30	0.40
$2 \to 3$	0.25	0.25	0.03	0.35	0.45
$3 \to 4$	0.20	0.30	0.04	0.40	0.50
$4 \to 5$	0.15	0.35	0.05	0.45	0.55

The thresholds are relaxed progressively at higher levels to account for the increased difficulty of balancing more balls simultaneously.

E Discussion

Architectural vs. augmentation guarantees. In our runs the failure of DSHC_OFF behaves as a budget-limited barrier rather than a convergence delay: slot identity becomes a deterministic proxy for curriculum stage and blocks promotion beyond k=2 for the full 30 000-iteration budget. PFDS avoids this by construction; without augmentation it advances to k=5 but is slower to reach k≥3 (∼16.9k vs. ∼3.9k iterations, Table 2). Whether longer training or different seeds would eventually unblock DSHC_OFF is left to future work.

Why does DSHC fail without augmentation in our runs? Beyond the symmetry-group argument, the failure has a concrete curriculum-dynamics signature. When slot-permutation augmentation is disabled, slot index i is consistently occupied by ball i across an episode; because the curriculum activates balls sequentially, the slot index becomes a deterministic proxy for the current curriculum stage. The policy keys on slot identity rather than physical state, and fails to satisfy the *conjunction* of promotion criteria at k=2: the episode-length ratio remains high (the agent does not fail outright) but the support margin never reaches the promotion threshold. PFDS is immune by construction, since its per-frame pooling discards slot-identity information. Figure 9 reports the resulting traces in the runs we observed.

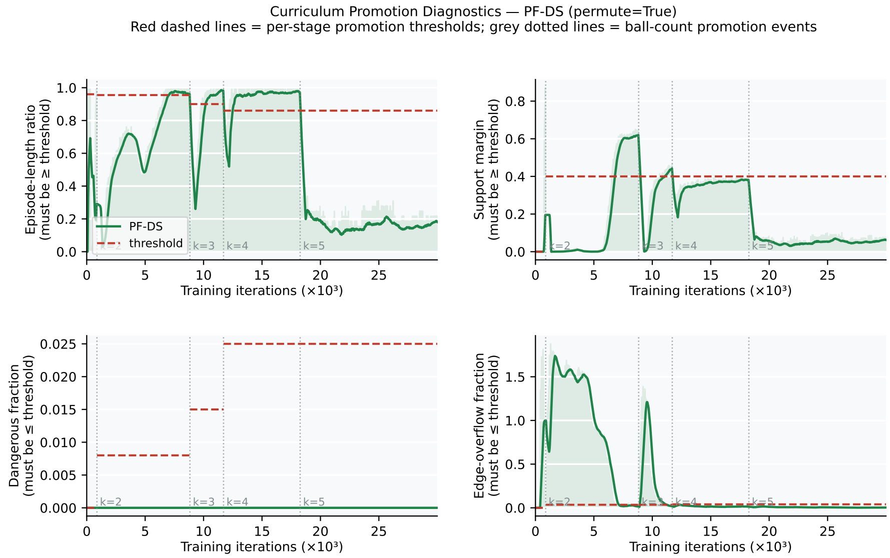

Figure 8: Curriculum promotion-criterion traces for a representative PFDS training run. Four of the six promotion conditions are shown: episode-length ratio, support margin, dangerous-state fraction, and edge-overflow fraction. Velocity- and angular-velocity-tracking errors are omitted for space but must also satisfy their thresholds. Vertical dashed lines mark the iterations at which the curriculum advances to the next level.

Why is PFDS slower without augmentation? Without slot-permutation augmentation, the training distribution samples only the identity element of G_{frame} at each step, reducing effective distributional diversity. Since PFDS is G_{frame}-invariant by construction, this does not prevent convergence but does increase the number of iterations required to cover the state space (Table 2: $\sim$16.9k vs. $\sim$3.9k iterations to $k \geq 3$).

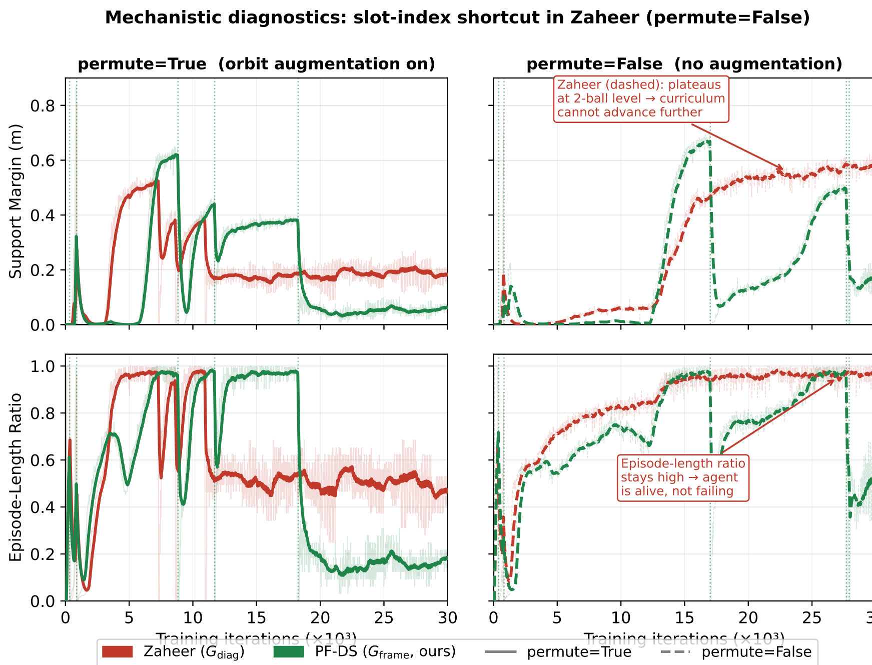

Figure 9: Curriculum-progress diagnostic for the slot-index shortcut. Rows show two promotion-related signals (top: support margin; bottom: episode-length ratio). Columns compare slot-permutation augmentation on (left) and off (right). DSHC (dashed) and PFDS (solid) are overlaid. Under augmentation-off, DSHC_OFF keeps the episode-length ratio high but cannot push the support margin to its promotion threshold, so the curriculum stalls at $k=2$; PFDS_OFF reaches both thresholds and advances. Under augmentation-on, both encoders meet the conjunction of promotion criteria.